Project Review I – Sample Submission

2. Azure Deployment Strategy – Detailed

Category	Detail
Architecture Model	Cloud Model: Public Cloud (Microsoft Azure) Service Model: PaaS for App Service, SQL DB; SaaS for Power BI Embedded. Hosting Region: Central India (Primary), South India (DR).
System Design	App Tier: Azure App Service Premium v2 (P1v2) Data Tier: Azure SQL Database (S2) & Blob Storage (Hot Tier, GRS) Network Tier: Azure Front Door + CDN for static content.
Reliability	High Availability: App Service SLA 99.95%, geo-replication for DB, GRS for Blob Storage. Disaster Recovery: SQL DB backups every 12h (7-day retention), Blob snapshots every 24h, DR failover in South India.
Security Measures	Authentication: Azure AD B2C with MFA. Authorization: RBAC for student/admin roles. Encryption: AES-256 at rest, TLS 1.2 in transit. Network Security: NSGs, private endpoints for DB & storage.
Operational Excellence	DevOps: Azure DevOps Pipelines integrated with GitHub. Deployment: Blue-Green via deployment slots. Monitoring: Azure Monitor & Application Insights with performance alerts.
Cost Optimization	Auto-scale App Service 1–3 instances based on CPU >70%. Blob Lifecycle: Move files >1 year old to Cool tier. Reserved Instance pricing for predictable workloads.

3. Infrastructure Requirements – Functional

Feature	Expected Usage	Azure Spec	Scaling Method

User Registration & Login	5,000 logins/day, 100 concurrent	App Service (P1v2), Azure AD B2C (10k MAUs)	Auto-scale 1–3 instances
Record Achievements	1M records/year	Azure SQL DB (S2, 50 DTUs, 250 GB)	Index optimization
Attach Certificates & Media	50MB/student/year × 5,000	Azure Blob Storage (Hot Tier, 1 TB)	Hot → Cool tier after 1 year
Generate Reports	100 concurrent views	Power BI Embedded (A1 SKU)	Dataset refresh 8x/day
Search & Filter	<1 sec response for 1M rows	SQL Full-Text Search	Index tuning
Admin Approval Workflow	~2,000 executions/month	Azure Functions (Consumption Plan)	Event-driven scaling

3. Infrastructure Requirements – Non-Functional

NFR	Metric Target	Azure Configuration	Reasoning
Scalability	Handle 3x load spikes	App Service Auto-scale 1–3 instances	Elastic scaling for exams/events
Performance	<500 ms API response	SQL indexing, caching	Smooth user experience
Availability	99.9% uptime	Geo-replication, GRS	Resiliency against failure
Security	MFA, TLS 1.2	Azure AD B2C, Private endpoints	Data protection
Cost	₹8,000/month cap	Azure Cost Management alerts	Budget control
Compliance	India data residency	Central India + DR in South India	Regulatory compliance

4. Azure Services Mapping – Expanded

Requirement	Azure Service	SKU/Tier	Purpose	Justification	Est. Monthly Cost (₹)
Web App Hosting	Azure App Service	Premium v2 (P1v2)	Host UI + APIs	High performance, SLA 99.95%, supports auto-scale	3,500
Database	Azure SQL Database	Standard S2	Achievement records	Scalable relational DB, automated backups	1,200

File Storage	Azure Blob Storage	Hot Tier, GRS	Certificates & media	High availability, redundancy	800
Authentication	Azure AD B2C	10k MAUs	Secure login	Supports MFA, RBAC, social logins	500
Workflow Automation	Azure Functions	Consumption Plan	Approval process	Event-driven, pay-per-use	200
Reporting	Power BI Embedded	A1 SKU	Dashboards/reports	Embedded analytics, no per-user license	1,000
Load Balancing	Azure Front Door	Standard	Traffic routing	Improves latency, HA	600
Monitoring	Azure Monitor + App Insights	Standard	Track performance	Centralized monitoring	400
Backup & DR	Azure Backup	Vault (250 GB)	SQL & Blob backup	DR compliance	300
CI/CD	Azure DevOps	Basic Plan	Deployment automation	Seamless GitHub integration	Free

5. High-Level Architecture Diagram

Diagram shows: End Users → Azure Front Door → App Service → SQL DB & Blob Storage → Azure Functions → Power BI Embedded; monitored via Azure Monitor & backed up via Azure Backup.